

7.2.5 Sponsor Responsibility for Reporting Adverse Events

All Adverse Events and Patient Events and Incidents will be reported to regulatory authorities, IRB/IECs and investigators in accordance with all applicable global laws and regulations, i.e., per ICH Topic E6 (R1) Guidelines for Good Clinical Practice.

7.3 TRIAL GOVERNANCE AND OVERSIGHT

7.3.1 Scientific Advisory Committee

This trial was developed in collaboration with a Scientific Advisory Committee (SAC). The SAC comprises both Sponsor and non-Sponsor scientific experts who provide input with respect to trial design, interpretation of trial results and subsequent peer-reviewed scientific publications.

7.3.2 Data Monitoring Committee

To supplement the routine trial monitoring outlined in this protocol, an external Data Monitoring Committee (DMC) will monitor the interim data from this trial. The voting members of the committee are external to the Sponsor. The members of the DMC must not be involved with the trial in any other way (e.g., they cannot be trial investigators) and must have no competing interests that could affect their roles with respect to the trial.

The DMC will make recommendations to the Sponsor per the DMC Charter regarding steps to ensure both subject safety and the continued ethical integrity of the trial. Also, the DMC will review interim trial results, consider the overall risk and benefit to trial participants (see Section 8.7 - Interim Analyses) and recommend to the Sponsor if the trial should continue in accordance with the protocol.

Specific details regarding composition, responsibilities, and governance, including the roles and responsibilities of the various members and the Sponsor protocol team; meeting facilitation; the trial governance structure; and requirements for and proper documentation of DMC reports, minutes, and recommendations will be described in a separate charter that is reviewed and approved by the DMC. The DMC will monitor the trial at an appropriate frequency, as described in the detailed DMC charter. The DMC will also make recommendations to the Sponsor executive committee and/or protocol team regarding steps to ensure both subject safety and the continued ethical integrity of the trial.

A DMC recommendation will be communicated to the Sponsor as agreed to in the DMC Charter.

8.0 STATISTICAL ANALYSIS PLAN

This section outlines the statistical analysis strategy and procedures for the study. If, after the study has begun, changes made to primary and/or key secondary hypotheses, or the statistical methods related to those hypotheses, then the protocol will be amended (consistent with ICH Guideline E-9). Changes to exploratory or other non-confirmatory analyses made after the protocol has been finalized, but prior to the conduct of any analysis, will be documented in a supplemental SAP (sSAP) and referenced in the Clinical Study

Report (CSR) for the study. Post hoc exploratory analyses will be clearly identified in the CSR.

8.1 Statistical Analysis Plan Summary

Key elements of the statistical analysis plan are summarized below; the comprehensive plan is provided in Sections 8.2 through 8.12.

Study Design Overview	A Phase 3 Study of Platinum+ Pemetrexed Chemotherapy with or without Pembrolizumab (MK-3475) in First Line Metastatic Nonsquamous Non-small Cell Lung Cancer Subjects (KEYNOTE-189)
Treatment Assignment	Approximately 570 subjects will be randomized in a 2:1 ratio to receive pembrolizumab or saline placebo in combination with pemetrexed/platinum. Stratification factors are in Section 5.4. This is a randomized double-blinded study.
Analysis Populations	Efficacy: Intention to Treat (ITT) Safety: All Subjects as Treated (ASaT)
Primary Endpoints/Hypotheses	• Progression-free Survival (PFS) per RECIST 1.1 assessed by BICR • Overall Survival (OS)
Statistical Methods for Key Efficacy Analyses	The primary hypotheses for PFS and OS will be evaluated by comparing pembrolizumab to saline placebo in combination with pemetrexed/platinum using a stratified Log-rank test. The HR will be estimated using a stratified Cox regression model. Event rates over time will be estimated within each treatment group using the Kaplan-Meier method.
Statistical Methods for Key Safety Analyses	The analysis of safety results will follow a tiered approach. There are no Tier 1 safety parameters in this trial. All safety parameters are considered either Tier 2 or Tier 3. Tier 2 parameters will be assessed via point estimates with 95% confidence intervals provided for between-group comparisons; only point estimates by treatment group are provided for Tier 3 safety parameters. The between-treatment difference will be analyzed using the Miettinen and Nurminen method. In the primary safety comparison, subjects who crossover to pembrolizumab are censored at time of crossover (ie, AEs occurring during treatment with pembrolizumab are excluded for control-arm subjects). An exploratory safety analysis will be conducted for the crossover population including all safety events starting from the date of first dose of pembrolizumab.
Interim Analyses	Two interim analyses are planned in this study. Results will be reviewed by an external data monitoring committee. Details are provided in Section 8.7. • Interim analysis 1 (IA1) ○ Timing: To be performed after target number of PFS events (~370) are observed ○ Purpose: To demonstrate superiority of pembrolizumab in combination with pemetrexed/platinum in PFS and OS. ORR will be tested after superiority of pembrolizumab in combination with pemetrexed/platinum is demonstrated in PFS and OS. • Interim analysis 2 (IA2) ○ Timing: To be performed after ~468 PFS events are observed ○ Purpose: To demonstrate superiority of pembrolizumab in combination with pemetrexed/platinum in PFS and OS.

Final Analysis	- Final analysis (FA) - Timing: To be performed after target number of deaths (~416) are observed - Purpose: To demonstrate superiority of pembrolizumab in combination with pemetrexed/platinum in OS.
Multiplicity	The overall Type I error rate for each endpoint in the group sequential tests is strictly controlled at 2.5% (one-sided); for both PFS and OS, this is based on the Lan-DeMets O'Brien-Fleming spending function (see Section 8.7 for details). Between the endpoints, the type I error is controlled by the following rollover rule. The total type I error allocated to PFS (0.0095) is subject to rollover to OS if the PFS test is positive. The type I error allocated to OS (0.0155) is subject to rollover to PFS if the OS test is positive. Furthermore, the total type I error (0.025) is subject to rollover to ORR at IA1 if the PFS and OS tests are both positive.
Sample Size and Power	Enrollment of 570 subjects is assumed to occur over 12 months at 2:1 ratio between the experimental and control groups. The actual enrollment is 616 subjects within 13 months. With 370 PFS events at IA1, the study has ~72% power for detecting a PFS HR of 0.7 at 0.0095 (one-sided) and ~84% power for detecting a HR of 0.7 at 0.025 (one-sided). With 468 PFS events at IA2, the study has ~90% power for detecting a HR of 0.7 at 0.0095 (one-sided) and ~96% power for detecting a HR of 0.7 at 0.025 (one-sided). The duration of PFS in the control group is assumed to follow an exponential distribution with a median of 6.5 months based on historical data. The assumed follow-up time after last patient enrolled is 13 months for IA2. An exponential dropout rate of 0.35% per month is assumed. With 242 deaths at IA1, the study has ~37% power for detecting an OS HR of 0.7 at 0.0155 (one-sided) and ~47% power for detecting a HR of 0.7 at 0.025 (one-sided) when the PFS test is significant. With 332 deaths at IA2, the study has ~73% power for detecting a HR of 0.7 at 0.0155 (one-sided) and ~80% power for detecting a HR of 0.7 at 0.025 (one-sided) when the PFS test is significant. With 416 deaths at FA, the study has ~90% power for detecting a HR of 0.7 at 0.0155 (one-sided) and ~93% power for detecting a HR of 0.7 at 0.025 (one-sided) when the PFS test is significant. The duration of OS in the control group is assumed to follow an exponential distribution with a median of 13 months based on historical data. The exponential dropout rate assumed for OS is 0.1% per month.

Following Amendment 09, the pre-planned second interim analysis (IA2) is no longer required and will not be performed. This is because of the study hypotheses for PFS, OS and ORR being supported at the first interim analysis (IA1) with data cutoff of November 8, 2017. The pre-specified FA at approximately 416 OS events will be performed without multiplicity adjustment. If median OS is not adequately assessed for the pembrolizumab combination arm at the pre-specified FA, an updated analysis may be conducted with a minimum of 30 months follow-up after the last patient randomized.

8.2 Responsibility for Analyses/In-House Blinding

The statistical analysis of the data obtained from this study will be the responsibility of the Clinical Biostatistics department of the Sponsor.

The Sponsor will generate the randomized allocation schedule(s) for study treatment assignment for this protocol, and the randomization will be implemented in IVRS.

This trial is double blinded with a crossover phase. At the time of documented progression, subjects will have treatment assignment unblinded and be able to continue therapy in the Crossover Phase, please refer to Section 2.1 Trial Design for details. In addition, independent central radiologist(s) will perform the central imaging review without knowledge of treatment assignment.

The study team at the Sponsor consisting of clinical, statistical, statistical programming and data management personnel, will be blinded to subject-level PD-L1 biomarker results. An unblinded data management personnel, unblinded Sponsor statistician and unblinded Sponsor statistical programmer will have access to the subject-level PD-L1 results for the purpose of data review and will have no other responsibilities associated with the study. A summary of PD-L1 biomarker prevalence may be provided to the study team at the Sponsor by the IVRS vendor or the unblinded Sponsor statistician.

An external data monitoring committee (eDMC) will be convened to review accumulating safety to provide an opportunity to terminate the study early if there are concerns regarding safety. The eDMC will also review the unblinded efficacy results at the planned interim analysis. The eDMC responsibilities and review schedules will be outlined in the eDMC charter. The recommendation of the eDMC will be communicated to an executive oversight committee of the Sponsor and, in the event of a recommendation to halt the trial early due to safety concerns, to the appropriate regulatory agencies. If the eDMC recommends modifications to the design of the protocol or discontinuation of the study, this executive oversight committee may be unblinded to results at the treatment level in order to act on these recommendations.

Limited numbers of additional Sponsor personnel may be unblinded, if required, in order to act on the recommendations of the eDMC. The extent to which individuals are unblinded with respect to the results will be documented. Additional logistical details, revisions to the above plan and data monitoring guidance will be provided in the eDMC Charter.

Positive results for both PFS and OS were observed at the FA. Thus a limited number of Sponsor personnel have been unblinded to support regulatory filing. The extent to which individuals were unblinded to the results has been documented.

After Amendment 10, the subject, investigator, and Sponsor personnel or delegate(s) who are involved in the treatment administration or clinical evaluation of the subjects will be made aware of the group assignments.

8.3 Hypotheses/Estimation

Objectives and hypotheses of the study are stated in Section 3.0.

8.4 Analysis Endpoints

8.4.1 Efficacy Endpoints

Primary

Progression-free survival (PFS) – RECIST 1.1 assessed by BICR

PFS is defined as the time from randomization to the first documented disease progression per RECIST 1.1 based on BICR or death due to any cause, whichever occurs first. See Section 8.6.1 for definition of censoring.

Overall Survival

OS is defined as the time from randomization to death due to any cause. Subjects without documented death at the time of analysis will be censored at the date of last known contact.

Secondary

Objective Response Rate (ORR) – RECIST 1.1 assessed by BICR

ORR is defined as the proportion of subjects who have a CR or a PR. Responses are based on confirmed assessments by BICR per RECIST 1.1.

Duration of Response (DOR) – RECIST 1.1 assessed by BICR

For subjects who demonstrated confirmed CR or PR, DOR is defined as the time from first documented evidence of CR or PR until disease progression or death. DOR for subjects who have not progressed or died at the time of analysis will be censored at the date of their last tumor assessment.

8.4.2 Safety Endpoints

Safety measurements are described in Section 7.

8.5 Analysis Populations

8.5.1 Efficacy Analysis Populations

The Intention-to-Treat (ITT) population will serve as the population for primary efficacy analysis. All randomized subjects will be included in this population. Subjects will be included in the treatment group to which they are randomized.

If an unexpectedly large number of randomized subjects are not treated, analyses may be performed using the Full Analysis Set (FAS), including all randomized subjects who received at least 1 dose of study treatment and did not have a major protocol violation.

Details on the approach to handling missing data are provided in Section 8.6 Statistical Methods.

8.5.2 Safety Analysis Populations

The All Subjects as Treated (ASaT) population will be used for the analysis of safety data in this study. The ASaT population consists of all randomized subjects who received at least one dose of study treatment. Subjects will be included in the treatment group corresponding to the study treatment they actually received for the analysis of safety data using the ASaT population. For most subjects this will be the treatment group to which they are randomized. Subjects who take incorrect study treatment for the entire treatment period will be included in the treatment group corresponding to the study treatment actually received. Any subject who receives the incorrect study medication for one cycle but receives the correct treatment for all other cycles will be analyzed according to the correct treatment group and a narrative will be provided for any events that occur during the cycle for which the subject is incorrectly dosed.

At least one laboratory or vital sign measurement obtained subsequent to at least one dose of study treatment is required for inclusion in the analysis of each specific parameter. To assess change from baseline, a baseline measurement is also required.

Details on the approach to handling missing data for safety analyses are provided in Section 8.6 Statistical Methods.

8.6 Statistical Methods

8.6.1 Statistical Methods for Efficacy Analyses

This section describes the statistical methods that address the primary and secondary objectives. Methods related to exploratory objectives will be described in the sSAP.

Efficacy results that will be deemed to be statistically significant after consideration of the Type I error control strategy are described in Section 8.8, Multiplicity. Nominal p-values will be computed for other efficacy analyses, but should be interpreted with caution due to potential issues of multiplicity.

All statistical tests, unless otherwise specified, will be stratified for treatment and stratification factors.

8.6.1.1 Progression-free Survival (PFS)

The non-parametric Kaplan-Meier method will be used to estimate the PFS curve in each treatment group. The treatment difference in PFS will be assessed by the stratified log-rank test (based on the stratification factors defined in Section 5.4). A stratified Cox proportional hazard model with Efron's method of tie handling will be used to assess the magnitude of the treatment difference (ie, hazard ratio) between the treatment arms. The hazard ratio and its 95% confidence interval from the stratified Cox model with Efron's method of tie handling and with a single treatment covariate will be reported. The stratification factors used for randomization (See Section 5.4) will be applied to both the stratified log-rank test and the stratified Cox model.

Since disease progression is assessed periodically, PD can occur any time in the time interval between the last assessment where PD was not documented and the assessment when PD is documented. For the primary analysis, for the subjects who have PD, the true date of disease

progression will be approximated by the date of the first assessment at which PD is objectively documented per RECIST 1.1 by BICR, regardless of discontinuation of study drug. Death is always considered as a confirmed PD event. Sensitivity analyses will be performed for comparison of PFS based on investigator's assessment.

In order to evaluate the robustness of the PFS endpoint per RECIST 1.1 by central imaging vendor, we will perform 2 sensitivity analyses with different sets of censoring rules. The first sensitivity analysis is the same as the primary analysis except that it censors at the last disease assessment without PD when PD or death is documented after more than one missed disease assessment. The second sensitivity analysis is the same as the primary analysis except that it considers discontinuation of treatment or initiation of an anticancer treatment subsequent to discontinuation of study-specified treatments, whichever occurs later, to be a PD event for subjects without documented PD or death. The censoring rules for primary and sensitivity analyses are summarized in [Table 11]. In case there is an imbalance between the treatment groups on disease assessment schedules or censoring patterns, we will perform an additional PFS sensitivity analysis using time from randomization to scheduled tumor assessment time instead of actual tumor assessment.

Table 11 Censoring Rules for Primary and Sensitivity Analyses of PFS

Situation	Primary Analysis	Sensitivity Analysis 1	Sensitivity Analysis 2
No PD and no death; new anticancer treatment is not initiated	Censored at last disease assessment	Censored at last disease assessment	Censored at last disease assessment if still on study therapy; progressed at treatment discontinuation otherwise
No PD and no death; new anticancer treatment is initiated	Censored at last disease assessment before new anticancer treatment	Censored at last disease assessment before new anticancer treatment	Progressed at date of new anticancer treatment
PD or death documented after ≤ 1 missed disease assessment	Progressed at date of documented PD or death	Progressed at date of documented PD or death	Progressed at date of documented PD or death
PD or death documented after ≥ 2 missed disease assessments	Progressed at date of documented PD or death	Censored at last disease assessment prior to the ≥ 2 missed disease assessment	Progressed at date of documented PD or death

In case the proportional hazards assumption is not valid, Restricted Mean Survival Time (RMST) method may be conducted for PFS to account for the possible non-proportional hazards effect.

An exploratory analysis of PFS2, defined as the time from randomization to second/subsequent disease progression after initiation of new anti-cancer therapy, or death from any cause, whichever first, will be carried out. Patients alive and for whom a second objective disease progression has not been observed will be censored at the last time known to be alive and without second objective disease progression.

Further details of sensitivity analyses will be described in sSAP as needed.

8.6.1.2 Overall Survival (OS)

The non-parametric Kaplan-Meier method will be used to estimate the survival curves. The treatment difference in survival will be assessed by the stratified log-rank test (based on the stratification factor defined in Section 5.4). A stratified Cox proportional hazard model with Efron's method of tie handling will be used to assess the magnitude of the treatment difference (ie, the hazard ratio). The hazard ratio and its 95% confidence interval from the stratified Cox model with a single treatment covariate will be reported. The stratification factors used for randomization (See Section 5.4) will be applied to both the stratified log-rank test and the stratified Cox model. Restricted Mean Survival Time (RMST) method may be conducted for OS to account for the possible non-proportional hazards effect.

Since subjects in the control arm are allowed to switch to the pembrolizumab treatment after progressive disease, adjustment for the effect of crossover on OS may be performed based on recognized methods, eg, a two-stage method or the Rank Preserving Structural Failure Time (RPSFT) model proposed by Robins and Tsiatis [43], based on an examination of the appropriateness of the data to the assumptions required by the methods.

Further details of sensitivity analyses will be described in sSAP as needed.

8.6.1.3 Objective Response Rate (ORR) and Duration of Response (DOR)

Stratified Miettinen and Nurminen's method will be used for comparison of the ORR between 2 treatment groups. The difference in ORR and its 95% confidence interval from the stratified Miettinen and Nurminen's method with strata weighting by sample size with a single treatment covariate will be reported. The stratification factors used for randomization (See Section 5.4) will be applied to the analysis.

If sample size permits, duration of response will be summarized descriptively using Kaplan-Meier medians and quartiles. Only the subset of patients who show a complete response or partial response will be included in this analysis.

[Table 12] summarizes the primary analysis approach for primary and key secondary efficacy endpoints. Sensitivity analysis methods are described above for each endpoint as applicable.

The strategy to address multiplicity issues with regard to multiple efficacy endpoints, and interim analyses is described in Section 8.7 Interim Analyses and in Section 8.8 Multiplicity.

Table 12 Analysis Strategy for Key Efficacy Endpoints

Endpoint/Variable (Description, Time Point)	Statistical Method [†]	Analysis Population	Missing Data Approach
Primary Endpoints			
PFS per RECIST 1.1 by central imaging vendor	<u>Test:</u> Stratified Log-rank test to assess the treatment difference <u>Estimation:</u> Stratified Cox model with Efron's tie handling method to assess the magnitude of treatment difference	ITT	• Primary censoring rule • Sensitivity analysis 1 • Sensitivity analysis 2
OS	<u>Test:</u> Stratified Log-rank test to assess the treatment difference <u>Estimation:</u> Stratified Cox model with Efron's tie handling method to assess the magnitude of treatment difference	ITT	Model based (censored at last known alive date)
Secondary Endpoint			
ORR per RECIST 1.1 by central imaging vendor	<u>Stratified M&N method with sample size weights^{††}</u>	ITT	Subjects without assessments are considered non-responders and conservatively included in denominator
[†] Statistical models are described in further detail in the text. For stratified analyses, the stratification factors used for randomization (Section 5.4) will be applied to the analysis. ^{††} Miettinen and Nurminen method			

8.6.2 Statistical Methods for Safety Analyses

Safety and tolerability will be assessed by clinical review of all relevant parameters including adverse experiences (AEs), laboratory tests, vital signs, etc.

Adverse Events

Adverse events (AEs) will be coded using the standard MedDRA and grouped system organ class. Adverse events (AEs) will be graded by the investigator according to the Common Terminology Criteria for Adverse Events (CTCAE), version 4.0.

Tiered Approach

The analysis of safety results will follow a tiered approach [Table 13]. The tiers differ with respect to the analyses that will be performed. "Tier 1" safety endpoints that will be subject to inferential testing for statistical significance with p-values and 95% confidence intervals provided for between-group comparisons. For this protocol, there are no Tier 1 events. Other safety parameters will be considered Tier 2 or Tier 3. Tier 2 parameters will be assessed via point estimates with 95% confidence intervals provided for between-group comparisons; only point estimates by treatment group are provided for Tier 3 safety parameters.

Adverse experiences (specific terms as well as system organ class terms) that are not pre-specified as Tier-1 endpoints will be classified as belonging to "Tier 2" or "Tier 3", based on the number of events observed. Membership in Tier 2 requires that at least 4 subjects in

any treatment group exhibit the event; all other adverse experiences and predefined limits of change will belong to Tier 3.

The threshold of at least 4 events was chosen because the 95% confidence interval for the between-group difference in percent incidence will always include zero when treatment groups of equal size each have less than 4 events and thus would add little to the interpretation of potentially meaningful differences. Because many 95% confidence intervals may be provided without adjustment for multiplicity, the confidence intervals should be regarded as a helpful descriptive measure to be used in review, not a formal method for assessing the statistical significance of the between-group differences in adverse experiences and predefined limits of change.

Continuous measures such as changes from baseline in laboratory, vital signs, that are not pre-specified as Tier-1 endpoints will be considered Tier 3 safety parameters. Summary statistics for baseline, on-treatment, and change from baseline values will be provided by treatment group in table format.

In addition, the broad clinical and laboratory AE categories consisting of the percentage of subjects with any AE, any drug related AE, any Grade 3-5 AE, any serious AE, any AE which is both drug-related and Grade 3-5, any AE which is both serious and drug-related, dose modification due to AE, and who discontinued due to an AE, and death will be considered Tier 2 endpoints. 95% confidence intervals (Tier 2) will be provided for between- treatment differences in the percentage of subjects with events; these analyses will be performed using the Miettinen and Nurminen method (1985), an unconditional, asymptotic method.

Table 13 Analysis Strategy for Safety Parameters

Safety Tier	Safety Endpoint [†]	95% CI for Treatment Comparison	Descriptive Statistics
Tier 2	Any AE	X	X
	Any Serious AE	X	X
	Any Grade 3-5 AE	X	X
	Any Drug-Related AE	X	X
	Any Serious and Drug-Related AE	X	X
	Any Grade 3-5 and Drug-Related AE	X	X
	Dose Modification due to AE	X	X
	Discontinuation due to AE	X	X
	Death		
	Specific AEs, SOCs, or PDLCS [‡] (incidence ≥ 4 of subjects in one of the treatment groups)	X	X
Tier 3	Specific AEs, SOCs or PDLCS [‡] (incidence < 4 of subjects in all of the treatment groups)		X
	Change from Baseline Results (Labs, ECGs, Vital Signs)		X

8.6.3 Summaries of Demographic and Baseline Characteristics and Other Analysis

The comparability of the treatment groups for each relevant characteristic will be assessed by the use of tables and/or graphs. No statistical hypothesis tests will be performed on these characteristics. The number and percentage of subjects randomized, and the primary reasons for discontinuation will be displayed. Demographic variables (eg, age, gender) and baseline characteristics will be summarized by treatment either by descriptive statistics or categorical tables.

8.7 Interim Analyses

Two interim efficacy analyses are planned in addition to the FA.

- The first interim analysis (IA1) will evaluate PFS and OS. It will be performed after enrollment is complete and approximately 370 PFS events have been observed, approximately 19 months after first patient enrolled. It is estimated that approximately 242 deaths will be observed. ORR will be tested at this interim analysis if both the PFS and OS test results are significant.
- The second interim analysis (IA2) is the FA for PFS and will be performed after approximately 468 PFS events have been observed, approximately 26 months after first patient enrolled. An interim analysis of OS will be performed; it is estimated that approximately 332 deaths will be observed.
- The FA will evaluate OS only and will be performed after approximately 416 deaths have been observed. It is estimated that this will occur approximately 35 months after enrollment begins.

The analyses planned, endpoints evaluated and drivers of the timing are summarized in [Table 14].

Table 14 Analyses Planned, Endpoints Evaluated, and Drivers of Timing

Analysis	Endpoint(s)	Timing
IA1	PFS; OS; ORR if both PFS and OS are positive	~370 PFS events (~242 OS events expected at this time)
IA2	PFS; OS	~468 PFS events (~332 OS events expected at this time)
FA	OS	~416 OS events

Decisions to stop the trial early will be based on DMC recommendations with review by the Executive Oversight Committee.

Type I error control for the efficacy analyses as well as efficacy bounds are described in the next section.

Following Amendment 09, the pre-planned second interim analysis (IA2) is no longer required and will not be performed. This is because of the study hypotheses for PFS, OS and

ORR being supported at the first interim analysis (IA1) with data cutoff of November 8, 2017. The pre-specified FA at approximately 416 OS events will be performed without multiplicity adjustment. If median OS is not adequately assessed for the pembrolizumab combination arm at the pre-specified FA, an updated analysis may be conducted with a minimum of 30 months follow-up after the last patient randomized.

8.8 Multiplicity

The trial uses the graphical method of Maurer and Bretz [44] to provide strong multiplicity control for multiple hypotheses as well as interim analyses.

Figure 2 shows the initial one-sided α -allocation for each hypothesis in the ellipse representing the hypothesis. The weights for reallocation from each hypothesis to the others are represented in the boxes on the lines connecting hypotheses. This is further explained below.

See Figure 2 for the type I error reallocation strategy for endpoints PFS, OS, and ORR.

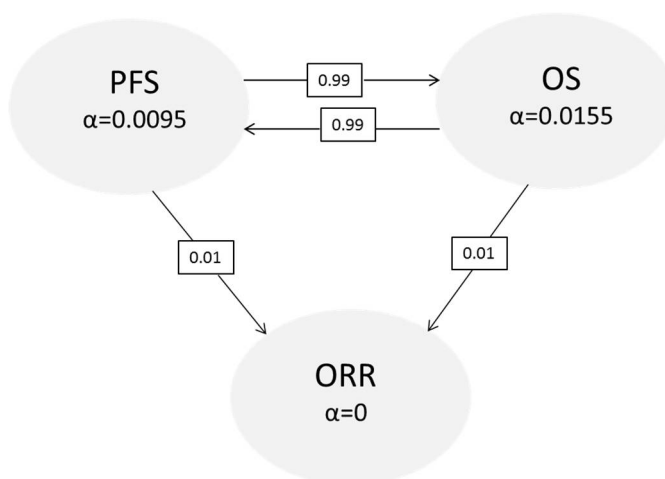


Figure 2 Type I Error Reallocation Strategy

PFS

PFS hypothesis will be tested at $\alpha=0.0095$. When OS test is significant, the PFS hypothesis may be tested at $\alpha=0.025$ (re-allocated α). The Lan-DeMets O'Brien-Fleming spending function was used to control the type I error in the interim analysis. [Table 15] below demonstrates the bounds and boundary properties for PFS hypothesis testing. The table will be updated using the actual number of PFS events at the interim and final PFS analyses.

Table 15 Boundary Properties for Planned Analyses of PFS Based on Potential Alpha-Levels to be Used for Testing

Analysis	Value	$\alpha=0.0095$	$\alpha=0.025$
IA 1: 79%	Z	2.6946	2.2676
N: 616	p (1-sided)	0.0035	0.0117
Events: 370	HR at bound	0.7427	0.7790
Month: 19.2	P(Cross) if HR=1	0.0035	0.0117
	P(Cross) if HR=0.7	0.7160	0.8439
IA2*	Z	2.3895	2.0222
N: 616	p (1-sided)	0.0084	0.0216
Events: 468	HR at bound	0.7910	0.8204
Month: 26	P(Cross) if HR=1	0.0095	0.0250
	P(Cross) if HR=0.7	0.9040	0.9550

*The FA of PFS will be performed at IA2.

If the OS superiority null hypotheses are rejected at an interim or FA, each PFS interim analysis test may be compared to its updated rejection boundary for formal testing.

OS

The OS hypothesis will be tested at $\alpha=0.0155$. When PFS test is significant, the OS hypothesis may be tested at $\alpha=0.025$ (re-allocated α). [Table 16] demonstrates the bounds and boundary properties for OS hypothesis testing. The HR of OS between the experimental group and control group is assumed to be 0.7. The table will be updated using the actual number of OS events at the interim and final OS analyses. The Lan-DeMets O'Brien-Fleming spending function was used to derive the design in α columns.

Table 16 Boundary Properties for Planned Analyses of OS

Analysis	Value	$\alpha=0.0155$	$\alpha=0.025$
IA 1: 58%	Z	2.9715	2.7215
N: 616	p (1-sided)	0.0015	0.0032
Events: 242	HR at bound	0.6662	0.6893
Month: 19.3	P(Cross) if HR=1	0.0015	0.0032
	P(Cross) if HR=0.7	0.3711	0.4695
IA 2: 80%	Z	2.4926	2.2831
N: 616	p (1-sided)	0.0063	0.0112
Events: 332	HR at bound	0.7482	0.7666
Month: 26.2	P(Cross) if HR=1	0.0068	0.0122
	P(Cross) if HR=0.7	0.7332	0.7991
Final	Z	2.2155	2.0300
N: 616	p (1-sided)	0.0134	0.0212
Events: 416	HR at bound	0.7940	0.8095
Month: 35	P(Cross) if HR=1	0.0155	0.0250
	P(Cross) if HR=0.7	0.9000	0.9300

A sensitivity analysis for OS will be pre-specified in the supplemental statistical analysis plan to account for any multiplicity concerns raised by not formally testing the OS null hypothesis such as DMC requests. This analysis will be performed if requested by regulators. However, there is no plan to do formal testing at these analyses or to declare a positive efficacy finding based on these results.

If the PFS superiority null hypotheses are rejected at any interim analyses, each OS interim and FA test may be compared to its updated rejection boundary for formal testing.

ORR

When both PFS and OS tests are significant, $\alpha=0.025$ will be allocated to the ORR test. If the OS test does not achieve statistical significance at IA1, the p value of the ORR test from IA1 will be compared to 2.5% if the null hypotheses for PFS and OS are rejected at a later time.

Following Amendment 09, the pre-planned second interim analysis (IA2) is no longer required and will not be performed. This is because of the study hypotheses for PFS, OS and ORR being supported at the first interim analysis (IA1) with data cutoff of November 8, 2017. The pre-specified FA at approximately 416 OS events will be performed without multiplicity adjustment. If median OS is not adequately assessed for the pembrolizumab combination arm at the pre-specified FA, an updated analysis may be conducted with a minimum of 30 months follow-up after the last patient randomized.

8.9 Sample Size and Power Calculations

This trial is well-powered for the primary and key secondary endpoints. Enrollment of 570 subjects is assumed to occur over 12 months at 2:1 ratio between the experimental and control groups. The actual enrollment is 616 subjects within 13 months. With 370 PFS events at IA1, the study has ~72% power for detecting a HR of 0.7 at 0.0095 (one-sided) and ~84% power for detecting a HR of 0.7 at 0.025 (one-sided). With 468 PFS events at IA2, the study has ~90% power for detecting a HR of 0.7 at 0.0095 (one-sided) and ~96% power for detecting a HR of 0.7 at 0.025 (one-sided). The duration of PFS in the control group is assumed to follow an exponential distribution with a median of 6.5 months based on historical data. The assumed follow-up time after last patient enrolled is 13 months for IA2. An exponential dropout rate of 0.35% per month is assumed.

With 242 deaths at IA1, the study has ~37% power for detecting a HR of 0.7 at 0.0155 (one-sided) and ~47% power for detecting a HR of 0.7 at 0.025 (one-sided) when the PFS test is significant. With 332 deaths at IA2, the study has ~73% power for detecting a HR of 0.7 at 0.0155 (one-sided) and ~80% power for detecting a HR of 0.7 at 0.025 (one-sided) when the PFS test is significant. With 416 deaths at FA, the study has ~90% power for detecting a HR of 0.7 at 0.0155 (one-sided) and ~93% power for detecting a HR of 0.7 at 0.025 (one-sided) when the PFS test is significant. The duration of OS in the control group is assumed to follow an exponential distribution with a median of 13 months based on historical data. The exponential dropout rate assumed for OS is 0.1% per month.

The sample size of 616 yields 89% ($\alpha=0.025$) power to detect an ORR difference from an underlying 45% response rate in the control group to 60% in the experimental group, when both PFS and OS tests are significant. The approximate treatment difference required to reach the bound (Δ ORR) is 0.0575.

Power and interim analysis calculations were performed using EAST 5 and the gsDesign R package.

Following Amendment 09, the pre-planned second interim analysis (IA2) is no longer required and will not be performed. This is because of the study hypotheses for PFS, OS, and ORR being supported at the first interim analysis (IA1) with data cutoff of November 8, 2017. The pre-specified FA at approximately 416 OS events will be performed without multiplicity adjustment. If median OS is not adequately assessed for the pembrolizumab combination arm at the pre-specified FA, an updated analysis may be conducted with a minimum of 30 months follow-up after the last patient randomized.

8.10 Subgroup Analyses and Effect of Baseline Factors

To determine whether the treatment effect is consistent across various subgroups, the between-group treatment effect for OS, PFS, and ORR (with a nominal 95% CI) will be estimated and plotted within each category of the following classification variables (a subgroup will not be analyzed if it includes <10% of the ITT population):

- Age category (<65, ≥65 years)
- ECOG Performance Scale (0, 1)
- Sex (female, male)
- Race (white, non-white)
- Geographic region (US, Ex US)
- Geographic region (EU, Ex EU)
- Smoking status (never, former/current)
- Brain metastasis status at baseline (yes, no)
- PD-L1 expression (unknown, TPS <1%, or TPS ≥1%)
- PD-L1 expression (unknown, TPS <50%, or TPS ≥50%)
- PD-L1 expression (unknown, TPS <1%, 1%≤TPS≤49%, or TPS ≥50%)
- Platinum chemotherapy (cisplatin, carboplatin)

8.11 Extent of Exposure

The extent of exposure will be summarized as duration of treatment in cycles.

9.0 LABELING, PACKAGING, STORAGE AND RETURN OF CLINICAL SUPPLIES

9.1 Investigational Product

The investigator shall take responsibility for and shall take all steps to maintain appropriate records and ensure appropriate supply, storage, handling, distribution and usage of investigational product in accordance with the protocol and any applicable laws and regulations.